ChatGPT

ZendoAI Roadmap – Axum, Rust ML, and UI Feature Integration (June 2025)

Phase 1: Immediate UI/UX Improvements

- **Image/Prediction Editor**
- Add in-browser image and prediction editing functionality
- Editable prediction textbox (display/update predictions)
- Ensure prediction edits are saved to backend
- **Gallery Upgrades**
- Batch upload (drag-and-drop multiple files)
- Image selection (checkboxes or multi-select)
- Batch deletion and label editing in gallery
- Visual feedback (toasts, loading spinners, error states)
- **Other UI Enhancements**
- Improved gallery layout and accessibility
- (Optional) Theme switcher, performance tuning

Phase 2: Backend Migration and Rust ML Integration

- **Axum Migration**
- Port API routes/middleware from FastAPI to Axum
- Refactor models, DB access, and image handling into Rust
- Maintain seamless REST/GraphQL API for frontend
- **Rust ML Core**
- Start with inference: Port CLIP/OpenCLIP prediction logic to Rust
- Prepare for future full-Rust ML training pipelines
- **Benchmarking**
- Compare FastAPI vs Axum for inference and image transforms
- Move all perf-critical code to Rust

Phase 3: Extended ML Features and Advanced Functionality

- **Batch Predictions & Evaluation**
- Batch image prediction and bulk labeling
- Log/track prediction metadata for performance analysis
- **Future ML: Training & Generative Models**
- Start building Rust-based model training (supervised, contrastive, etc.)
- Long-term: Add Rust-native generative models (GANs, diffusion, midi-to-audio, etc.)

Meta & Ops

- Keep README/docs updated as features migrate
- Set up robust CI/CD for new Rust backend
- Focus on features with direct portfolio/job search value (UI/UX and Rust ML advancements)